The number of points on the curve $y = \dfrac{(x+1)(3x-1)}{x-2}$ where the tangent line is horizontal is

If α, β and γ are constants such that the function
$f(x) = \begin{cases} \alpha x^2 + \beta x + \gamma & x < 0 \\ 2\cos x + \tan x & x \geq 0 \end{cases}$
is differentiable everywhere up to the second order, then $f(-1)$ is equal to

If $y = t^2 - t + 2$ and $x = t^3 + t$, then the value of $\frac{dy}{dx}|_{t=1}$ is equal to

What is the slope of the tangent line to the curve $x + \frac{\sqrt{2}}{4}\cos(xy) = y^2 + \frac{\pi}{2}$ at the point $\left(\frac{\pi}{2}, \frac{1}{2}\right)$

Find $\lim_{x\to 0}(1+3x)^{\frac{1}{2x}} =$

Find the derivative of $y = (\sin x)^{\csc^3 x}$ at $x = \dfrac{\pi}{6}$.
[Hint: Use Logarithmic differentiation]

A car A is travelling east at 68 km/h and a car B is travelling south at 80 km/h. Both are headed towards the same intersection of the roads. At what velocity are the cars approaching each other when car A is 500 m and car B is 1200m from the intersection?

Given $f(x) = (1+2x)^{5.5}e^{-4.4x}$, then using *linearization* of $f(x)$ at $x = 0$, an approximate value of $f(0.1)$ is

If $f(x) = \cot x \cdot \csc x$, then $f'\left(\frac{\pi}{4}\right) =$

Consider the equation $(y^2-1)\,x^2 + y^3 = \frac{1}{x}$. What is the value of y' at $y = -1$?

$$\lim_{\theta \to 0} \frac{\sin(2\theta) - \cos(5\,\theta) + \tan(3\,\theta) + 1}{\sin(3\,\theta) + \cos(6\,\theta) - \tan(2\theta) - 1} =$$

The equation of the normal line to the curve $f(x) = \dfrac{2 - x\,e^x}{3x + 1}$ at the point $(0, 2)$ is given by:

(8-points) Use continuity to evaluate the limit

$$\lim_{x \to -3} \arctan\left(\frac{x^2 + 7x + 12}{x^2 + 5x + 6}\right).$$

The vertical asymptote(s) of $f(x) = \dfrac{x^2 - x - 2}{x^2 - 4}$ is(are):

Let $f(x) = 2 - 4x$. The largest value of $\delta > 0$, such that if $0 < |x - 2| < \delta$ then $|f(x) + 6| < 0.12$, equals to

(a) 0.03
(b) 0.02
(c) 0.012
(d) 0.04
(e) 0.05

$$\lim_{x \to 1^+} \frac{|1 - x^2| - (x^3 - 1)}{(x-1)^2} =$$

Let $f(x)$ be a continuous function on the interval $[0, 4]$, then the expression

$$\frac{f(2.5) - f(0.5)}{2}$$

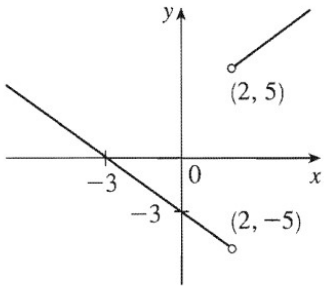
is

(a) the average rate of change of f over the interval $[0.5, 2.5]$
(b) the derivative of f at $t = 0.5$
(c) the slope of the tangent line at $t = 2$
(d) the slope of the tangent line at $t = 2.5$
(e) the average rate of change of f over the interval $[0, 4]$

The graph of the function f is given below, then

$f'(0) + \lim_{x \to 2^+} 4f(x) =$

(a) 19
(b) 14
(c) 3
(d) 5
(e) 12



Let $f(x)$ be a function such that $f(1) = 1$ and $\lim_{x \to 1} \frac{f(x) - 1}{x - 1} = \frac{1}{2}$. The equation of the tangent line to the graph of $f(x)$ at $x = 1$ is given by

If the line $y = 3x - 2$ is tangent to the parabola $y = ax^2 + bx$ at the point $(1, 1)$, then $a - b =$

6. (10 pts.) For what values of the constants c and d is the function solutions.

$$f(x) = \begin{cases} cx + d & x \leq 3 \\ dx^2 - 4 & x > 3 \end{cases}$$

continuous and differentiable on $\mathbb{R} = (-\infty, \infty)$?

) $\lim_{x \to 0} \dfrac{3^{(2+x)^2} - 81}{x}$

***(12 pts)* 2.**
Find an equation of the tangent line to the curve $x^y = y^x$ at the point $(2, 4)$.

***(3+3+6+6+6 pts)* 7.** Let $f(x) = (x-3)^2 e^x$.

a) Find the domain and intercepts of f

b) Find all vertical and horizontal asymptotes of the graph of f, if there are any. Explain.

c) Find the intervals where f is increasing/decreasing.

d) Find the intervals where f is concave up/down and all inflection points of f.